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Method of Manufacturing Piezoelectric Microactuators Having Wrap-Around Electrodes

Abstract

A method of manufacturing a piezoelectric microactuator having a wrap-around electrode includes forming a piezoelectric element having a large central electrode on a top face, and having a wrap-around electrode that includes the bottom face, two opposing ends of the device, and two opposing end portions of the top face. The device is then cut through the middle, separating the device into two separate piezoelectric microactuators each having a wrap-around electrode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-part of U.S. patent application Ser. No. 17/576,420 filed Jan. 14, 2022, which is a continuation of U.S. patent application Ser. No. 15/912,042 filed Mar. 5, 2018, now U.S. Pat. No. 11,227,629, the disclosures of which are incorporated by reference as if fully set forth herein.

FIELD OF THE INVENTION

[0002] This invention relates to the field of suspensions for hard disk drives. More particularly, this invention relates to a system and method of manufacturing suspension components or flexible circuits for use in disk drive suspensions.

BACKGROUND

[0003] Magnetic hard disk drives and other types of spinning media drives such as optical disk drives are well known. FIG. 1 is an oblique view of an exemplary prior art hard disk drive and suspension for which the present invention is applicable. The prior art disk drive unit **100** includes a spinning magnetic disk **101** containing a pattern of magnetic ones and zeroes on it that constitutes the data stored on the disk drive. The magnetic disk is driven by a drive motor (not shown). Disk drive unit **100** further includes a disk drive suspension **105** to which a magnetic head slider (not shown) is mounted proximate a distal end of load beam **107**. Suspension **105** is coupled to an actuator arm **103**, which in turn is coupled to a voice coil motor **112** that moves the suspension **105** arcuately in order to position the head slider over the correct data track on data disk **101**. The head slider is carried on a gimbal which allows the slider to pitch and roll so that it follows the proper data track on the disk, allowing for such variations as vibrations of the disk, inertial events such as bumping, and irregularities in the disk's surface.

[0004] Both single stage actuated disk drive suspensions and dual stage actuated (DSA) suspension are known. In a single stage actuated suspension, only voice coil motor **112** moves suspension **105**.

[0005] In a DSA suspension, as for example in U.S. Pat. No. 7,459,835 issued to Mei et al. as well as many others, in addition to voice coil motor **112** which moves the entire suspension, at least one microactuator is located on the suspension in order to effect fine movements of the magnetic head slider to keep it properly aligned over the data track on the spinning disk. The microactuator(s) provide much finer control and much higher bandwidth of the servo control loop than does the voice coil motor alone, which effects relatively coarse movements of the suspension and hence the magnetic head slider. A piezoelectric element or component, made of piezoelectric material, sometimes referred to simply as a PZT, is often used as the microactuator motor, although other types of microactuator motors are possible. In the discussion that follows, for simplicity the microactuator will be referred to simply as a "PZT," although it will be understood that the microactuator need not be of the PZT type.

[0006] FIG. 2 is a top plan view of the prior art suspension **105** in FIG. 1. Two PZT microactuators **14** are affixed to suspension **105** on microactuator mounting shelves **18** that are formed within base plate **12**, such that the PZTs span respective gaps in base plate **12**. Microactuators **14** are affixed to

mounting shelves **18** by non-conductive epoxy **16** at each end of the microactuators. The positive and negative electrical connections can be made from the PZTs to the suspension's flexible wiring trace and/or to the grounded base plate by a variety of techniques including those disclosed in commonly owned U.S. Pat. No. 7,751,153 to Kulangara et al.

[0007] In assembling a DSA suspension, the process typically includes the steps of: dispensing liquid adhesive such as epoxy onto the suspension and/or the PZT; positioning the PZT into place on the suspension; and curing the adhesive, typically by thermal curing, ultraviolet ("UV") curing, or other curing methods depending on the adhesive used. DSA suspensions often include both conductive epoxies and/or non-conductive epoxies to bond the PZT to the suspension. Conductive adhesives, such as silver-containing epoxies, are well known and are commonly used.

[0008] FIG. **3** shows a prior technique for bonding two PZTs in a DSA suspension including the electrical connection therebetween. FIG. **3** is not admitted as being "prior art" within the legal meaning of that term. Similarly, the processes described herein as applicable to FIG. **3** are also not admitted as being "prior art" within the legal meaning of that term. FIGS. **4** and **5** are cross sectional views taken along section lines C-C' and D-D' in FIG. **3**, respectively, showing the details of the bonding. As best seen in FIG. **4**, the PZT **330** has electrodes on both sides where the bottom electrode is grounded at one end by conductive epoxy **324** through gold **326** on grounded stainless steel **328** layer, and insulated at the other end by non-conductive epoxy **320**. As best seen in FIG. **5**, the top electrode is connected to a copper electrical contact pad **316** which is part of the suspension's electrical interconnect or flexible circuit and is insulated from the stainless steel substrate **312** by insulating layer **314** such as polyimide, by conductive epoxy **322** over non-conductive epoxy **320**. Electrical contact pad **316** provides the driving voltage for PZT **330**. Non-conductive epoxy **320** is primarily responsible for the mechanical bond between PZT **330** and stainless steel substrate **312**. In general, the height of the conductive epoxy **322** for the top electrode is difficult to control, and the overall PZT attachment process requires three epoxy bonding steps, which is time consuming and costly. Also, two separate curing steps are required for the epoxy on the bottom electrode and on the top electrode.

[0009] There are drawbacks to the prior methods of bonding PZTs to suspensions. It can be difficult to control exactly how much epoxy is dispensed, where the adhesive ends up due to flow of the liquid adhesive, and other issues. Various solutions have been proposed that involve, for example, channels underneath the PZTs to control the flow of adhesive and to channel any excess liquid epoxy away from sensitive areas. U.S. Pat. No. 6,856,075 to Houk, for example, proposes an adhesive attachment that has one or more reliefs under or partially under or adjacent to a PZT transducer to control the flow of adhesive by limiting or influencing adhesive travel or flow and simultaneously preventing excessive adhesive fillet height adjacent the piezoelectric motor. Additionally, if the PZT is located at or near the gimbal which carries the magnetoresistive read/write head, it becomes critical to be able to predict and control the flow of adhesive because differences in adhesive flow and distribution from one part to another can adversely affect the geometries, mechanical properties, and resulting performance of the suspension. These issues are particularly pronounced when the PZT is located at a particularly sensitive part of the suspension such as near or at the gimballed head slider. Repeatability and predictability are especially critical in that area. Still further, the presence of liquid epoxy and its dispensing equipment within the final assembly room represents both a potential source of contamination, as well as an additional and expensive manufacturing step.

[0010] Another drawback to the prior attachment means is the delays in assembly time required for multiple rounds of epoxy, including both conductive epoxy and non-conductive epoxy, to be dispensed and then cured. FIG. **6** illustrates the typical manufacturing process for PZT attachment, in which liquid epoxy is applied to the PZT, the interconnect, and/or the load beam before attaching the PZT. Conductive epoxy is dispensed at the interconnect (**610**), then non-conductive epoxy is dispensed at the load beam or other location for the PZT (**612**). The PZT is attached to the

suspension (614), and the epoxy is then cured (616) in a first curing step. Next, conductive epoxy is dispensed again (618), and then cured (620) in a second curing step.

SUMMARY

[0011] In order to address the foregoing disadvantages and other disadvantages of prior assembly processes, the present invention employs other types of adhesive than those traditionally used for DSA suspensions, and bonding and curing steps other than those traditionally used for DSA suspensions.

[0012] In one aspect, the invention employs adhesive films between the PZT and the suspension, and/or employs partially curing (B-staging) of a liquid or paste adhesive such as epoxy on the PZT before the PZT is placed onto the suspension component. The suspension component to which the PZT is affixed can be a base plate such as shown in FIGS. 1 and 2 for a baseplate mounted microactuator, a flexure such as shown in FIGS. 3-5 for a flexure mounted actuator, the suspension's load beam, the head slider itself in the case of a collocated PZT, or any other suspension component to which a microactuator motor may be mounted. After the two parts have been mated, the epoxy or other adhesive is finally cured. The curing can be thermal curing, UV curing, or other method of curing.

[0013] In a second aspect of the invention, the invention is of a method for producing a PZT microactuator or other electronic device having a wrap-around electrode, and of applying and using such a device. The wrap-around electrode is a conductive coating that wraps around at least part of the PZT to cover more than one face of the PZT, and thus conduct electricity to the opposite face. The wrap-around electrode simplifies both the assembly process and the final electrical connection(s) to the PZT in the completed suspensions. According to the method, a central electrode is first formed such as by sputtering on a first face of a wafer of piezoelectric material. A first side electrode is then formed on a first side or end of the wafer and over the adjacent edge, such that the first electrode extends onto the first face but is electrically not connected with, i.e., is discontinuous from, the central electrode on that face. Similarly, a second side electrode is also formed on a second side or end that is opposite the first side, with the second electrode also extending over an adjacent edge, such that the second electrode extends onto the first face but is also electrically discontinuous from the central. Conductive material is then deposited such as by sputtering on the second face of the wafer opposite the first face, with that conductive material extending to and being in electrical contact with the first and second side electrodes. The wafer now has one central electrode on the first face that covers most of the first face, and two side electrodes, with each side electrode covering not only its respective side but wrapping around that side and covering at least respective parts of the first and second faces, preferably at narrow respective strips on the first face on either side of the central electrode. The wafer is then cut in half. The result is two piezoelectric devices, each device having a wrap-around electrode such that the first face includes both electrodes. Both the drive and the ground connection can therefore both be made to the first face of the PZT, thus simplifying the electrical connections to it.

[0014] In addition, a system for printing conductive and non-conductive adhesives onto suspension components or flexible circuits is provided. The system includes a machine base, at least one suspension component positioned on the machine base, and a printing device. The printing device includes an aperture. The aperture is positioned over a predetermined location on the suspension component where adhesive material is to be deposited. The system also includes an application device configured to move in a shearing direction about the printing device and deposit the adhesive material into the aperture.

[0015] In some embodiments, the system can include a panel of a plurality of suspension components. Each of the suspension components in the panel include a predetermined location where the adhesive material is to be deposited. In some embodiments, the printing device comprises either a screen or a stencil. The adhesive material can include conductive adhesive material and/or non-conductive adhesive material. The adhesive material is a viscous fluid. In some

embodiments, the system can include a heat source configured to cure the adhesive material deposited into the aperture. In alternative embodiments, the system includes a UV light source configured to cure the adhesive material deposited into the aperture. In some embodiments, the adhesive is deposited on the suspension component at a location that imparts a structural effect on the suspension component. For example, the adhesive deposited on the suspension component may be used to increase the torsional stiffness of the suspension component.

[0016] In another aspect, a method of printing adhesive on a suspension component is provided. More specifically, one or more suspension components are loaded onto a machine base. A screen or stencil is placed over the one or more suspension components, the screen or stencil includes one or more apertures formed therein. An adhesive material is applied to the screen or stencil at a first location. A squeegee is positioned at the first location with the adhesive material and then the squeegee is moved across the screen or stencil to deposit the adhesive material into the one or more apertures. The adhesive material deposited into the aperture may then be partially cured.

[0017] The invention simplifies the assembly process for a DSA suspension and eliminates contamination sources in the sensitive final suspension assembly.

[0018] Exemplary embodiments of the invention will be further described below with reference to the drawings, in which like numbers refer to like parts. The drawing figures might not be to scale, and certain components may be shown in generalized or schematic form and identified by commercial designations in the interest of clarity and conciseness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a perspective view of a prior art disk drive assembly having a DSA suspension.

[0020] FIG. 2 is a top plan view of the prior art suspension 105 in FIG. 1.

[0021] FIG. 3 is a perspective view of a prior DSA suspension to which the present invention is applicable.

[0022] FIG. 4 is a sectional view taken along section line C-C' in FIG. 3.

[0023] FIG. 5 is a sectional view taken along section line D-D' in FIG. 3.

[0024] FIG. 6 is a process flow diagram of the process used to attach the PZT to the suspension in FIG. 3.

[0025] FIG. 7 is a process flow diagram of the simplified process used to attach a PZT to a suspension according to the invention.

[0026] FIG. 8 is a perspective view of a DSA suspension minus the flexure gimbal assembly, according to an embodiment of the present invention that employs integrated adhesive patterns printed onto the PZT.

[0027] FIG. 9 is an exploded view of the suspension of FIG. 8.

[0028] FIG. 10 is a perspective view of one of the PZTs of the suspension of FIG. 8, showing the integrated adhesive film underneath.

[0029] FIG. 11 is a perspective view of a portion of a DSA suspension according to an additional embodiment of the invention, in which integrated adhesive film is used on the top surface of the PZT.

[0030] FIG. 12 is a perspective view of the PZT in FIG. 11.

[0031] FIG. 13 is a side cut-away view of a PZT having a wrap-around electrode according to an additional embodiment of the invention.

[0032] FIG. 14 is a process flow diagram for the prior epoxy dispensing and bonding steps for the PZT of FIG. 3.

[0033] FIG. 15 is a process flow diagram for the prior epoxy dispensing and bonding steps and for the PZT of the present invention shown in FIG. 13.

[0034] FIGS. **16A-16G** illustrate the manufacturing steps for making the PZT with wrap-around electrode shown in FIG. **13**.

[0035] FIGS. **17A** and **17B** are isometric views of a PZT wafer at selected steps during the manufacturing process according to the present invention.

[0036] FIGS. **18A-18G** show an alternative manufacturing process that could be used to produce the PZT of FIG. **13**.

[0037] FIG. **19** is a perspective view of a PZT with adhesive being applied thereto according to the invention.

[0038] FIG. **20** is a top perspective view of the PZT of FIG. **19** illustrating the adhesives thereon being B-staged.

[0039] FIG. **21** is a side cutaway view of the PZT of FIG. **20** after it has been applied to a suspension.

[0040] FIG. **22** is a side cutaway view of the PZT having a wrap-around electrode after it has been applied to a suspension.

[0041] FIG. **23** is a perspective view showing a suspension, in accordance with an embodiment of the disclosure.

[0042] FIG. **24** illustrates a system for screen printing conductive and non-conductive adhesives onto suspension components or flexible circuits.

[0043] FIG. **25A** is a process flow diagram of a method used for screen printing conductive and non-conductive adhesives onto flexible circuits according to some examples.

[0044] FIG. **25B** is a process flow diagram of a method used for screen printing conductive and non-conductive adhesives onto suspension components according to some examples.

[0045] FIGS. **26A** and **26B** are top and perspective views, respectively, showing a printed adhesive formed on suspension components such as a stiffener or baseplate according to some examples.

[0046] FIGS. **27A** and **27B** are perspective views showing a printed adhesive formed on suspension components such as an integrated baseplate and load beam with hub geometry according to other examples.

[0047] FIG. **28** shows a perspective view of a suspension component with suspension arm having additional printed adhesive area on the suspension component.

DETAILED DESCRIPTION

[0048] A first aspect of the invention is the use of adhesive film to attach the PZT to the suspension. FIG. **7** illustrates the method. Conductive epoxy is dispensed at the interconnect (**710**). The PZT having the integrated adhesive film or other B-staged adhesive is then attached to the suspension (**712**), and the adhesive is then cured (**714**). Conductive epoxy is then dispensed at (**716**) and cured (**718**). The step of epoxy dispensing on the load beam is eliminated by using a PZT with integrated adhesive film. PZT with integrated adhesive film can be manufactured with either laminated adhesive film on it at the PZT wafer level or by printing or by a wafer backside coating process as used in the semiconductor industry. Because the adhesive film is attached to the PZT at the wafer level before dicing into individual PZT dies, process simplification and cost savings can be achieved. Also, tight control of the adhesive thickness can be achieved. As shown in FIG. **7** the use of a PZT with integrated adhesive also simplifies the process when compared to FIG. **6**. The step of non-conductive epoxy dispensing at the load beam is eliminated using the method of the invention.

[0049] A suspension design that facilitates the use of integrated adhesive film is shown in FIGS. **8-10**. FIG. **8** is a perspective view of a DSA suspension, less the flexure gimbal assembly for clarity of illustration, according to an embodiment of the present invention that employs integrated adhesive film or patterns printed onto the PZT. PZTs **830** are affixed to the suspension component to which it is mounted, in this case to baseplate **812**. Actuation of the PZTs moves load beam **820** such that the head slider which is located at the distal end of the load beam moves radially.

[0050] FIG. **9** is an exploded view of the suspension of FIG. **8**.

[0051] FIG. **10** is a perspective view of one of the PZTs **830** of suspension **810** of FIG. **8**, showing the integrated adhesive film or printed adhesive **834**, **836** underneath. In this embodiment, the integrated adhesive films **834**, **836** located at both ends of PZT **830** are formed by applying adhesive film or printing adhesive patterns on the PZT wafers before dicing and singulation into individual PZT pieces. The exposed PZT surface between the adhesives will allow for electrical connection with liquid epoxy to the suspensions's electrical circuit.

[0052] FIG. **11** is perspective view of a portion of a DSA suspension according to an additional embodiment of the invention, in which integrated adhesive film is used on the top surface of PZTs **1130** in order to bond the PZTs to recessed portions of baseplate **1112**.

[0053] FIG. **12** is a perspective view of the PZT in FIG. **11**. In this design, the integrated adhesive film or printed adhesive **1134** is located at the top surface of the PZT. The integrated adhesive film covers the entire top surface, or substantially the entire top surface, and thus no patterning of the adhesive is required. Also, conductive adhesive film can be used in this design, such that the grounding through the base plate can be achieved once the PZT is attached to the base plate. This will further eliminate the required step of grounding the PZT to the baseplate with conductive liquid epoxy as shown in FIG. **6**.

[0054] The adhesive film used can be either conductive or non-conductive, depending on whether an electrically conductive connection to the suspension or the interconnect circuit is desired, or a non-conductive connection to the suspension. Film adhesives are generally “preformed” or “B-staged,” and are available in rolls, sheets, or die-cut shapes.

B-Staged Epoxy

[0055] In a slightly different embodiment, instead of applying adhesive film to the PZT and/or to the suspension, adhesive is applied to the PZT and is B-staged before final assembly.

[0056] The term “B-staged” or “B-staging” as used herein means, after a flowable adhesive has been dispensed, partially hardening the adhesive so that its flow rate is substantially reduced to the point that it no longer flows freely as a liquid, but is not so hard such that it is no longer available for effectively adhering to another surface. B-staging involves temporarily exposing the adhesive to an environment which causes accelerated hardening of the adhesive, then removing the adhesive from that environment such that the hardening rate slows down considerably so that the adhesive does not substantially harden during assembly. The removal of the PZT from that increased hardening environment can include simply removing the hardening accelerant from the environment. B-staging can cure or otherwise harden the adhesive to a degree such that the adhesive is no longer tacky. One method of B-staging is to partially cure a cross-linking polymer such as epoxy, such as by applying heat and/or UV, such that the epoxy achieves less than 10% cross-linking, then removing the curing source. For epoxies that are B-staged using heat, the epoxy may be immediately quenched down to a lower temperature at which cross-linking is negligible, i.e., at which the epoxy effectively ceases to harden, in order to stop the cross-linking process. For epoxies that are B-staged using UV, removing the PZT from the increased hardening environment can mean simply turning off the UV curing lamps.

[0057] With some adhesives, the adhesive may be mixed into a solvent to form a slurry, the solvent being one that evaporates at a lower temperature than which cross-linking begins to occur significantly. The adhesive may be a printable paste that is applied to the PZT. After dispensing, the adhesive is exposed to a specified thermal regime designed to evolve a majority of the solvent from the material without significantly advancing resin cross-linking. The result is an epoxy or other adhesive that no longer flows, but that is still available for adhering to another surface with the full or nearly full adherent strength of the epoxy.

[0058] B-staging an adhesive permits the adhesive and substrate construction to be “staged,” or held for a period of time prior to the bonding and curing, without forfeiting performance. A secondary thermal cure cycle yields fully crosslinked, void-free bonds. As used herein, the term “fully crosslinked” means at least 90% crosslinked.

[0059] The adhesive may take the form of a solid, thermosetting paste. The adhesive may be a printable paste that is printed by any known printing techniques that are suitable for use with adhesive, including screen printing, stencil printing, ink jet printing, spraying, stamping, and others. An advantage of using such printing techniques is that the adhesive can be dispensed in very fine and precise patterns onto the PZT, which helps to achieve control and repeatability of the adhesive's total mass and distribution within the finished suspension. One commercially available silver-filled conductive epoxy that is suitable for fluid jetting, screen printing, and stamping is EPO-TEK® H20E by Epoxy Technology, Inc. of Billerica, Massachusetts.

[0060] A UV B-stage adhesive can be used. Such an adhesive is dispensed, then irradiated with UV energy in order to B-stage it. B-staging immediately after printing “freezes” the adhesives in position, which helps to precisely control any spread of the liquid epoxy. Unlike thermal staging, irradiating with UV energy eliminates the danger of advancing the thermoset reaction of the adhesive. UV B-staging can occur in seconds, while the thermal alternative can take an order of magnitude longer for the process.

[0061] Liquid epoxy or other adhesive may be first dispensed onto the PZT and/or onto the suspension, and then the epoxy is B-staged to the point that its flow is reduced to a negligible amount. The parts can then be assembled in the final, clean room assembly area for the disk drives, and the adhesive then fully cured either by heat or by UV. Such techniques have been used, or have been proposed to be used, in the integrated circuit (IC) packaging field under the broad term of wafer backside coating (WBC). Wafer backside coating techniques using both conductive and non-conductive adhesives can be adapted from die attach processes used in IC packaging to PZT attach processes for suspensions. Inkjet printing of polymers, both conductive and non-conductive, has also been proposed. Such inkjet printing techniques can be adapted for use in printing adhesives onto the PZTs for bonding those PZTs to suspensions.

[0062] It is anticipated that one method of production will be to begin with a wafer of PZT material, either applying already B-staged adhesive to it such as in adhesive film form or applying adhesive to it then B-staging the adhesive, then dicing the wafer into individual PZT microactuator motors. Pick-and-place machinery will be used to pick up the individual PZT die with the B-staged adhesive on it, assemble the PZT die to the suspension, and dwell there for the appropriate time and under the appropriate temperature and pressure conditions in order to fully cure the adhesive, and thus fully adhere the PZT to the suspension.

Wrap-Around Electrode

[0063] In another aspect, the invention is of a method of producing a piezoelectric microactuator or other electronic device having a wrap-around electrode, such that both the drive voltage and ground electrodes are located and accessible on the same side of the device.

[0064] FIG. **13** is a side cut-away view of a PZT according to an additional aspect of the invention. PZT **1330** has a bottom electrode **1334** and a “wrap-around” electrode **1336** that enables simplified electrical connections to the PZT. In this embodiment, the top electrode **1336** is wrapped around the PZT onto one end of the bottom PZT surface. The driving voltage is thus applied to the bottom surface of the PZT on a first end thereof, and the electrical ground is connected to the bottom surface on the opposite end. Thus, both electrodes **1334**, **1336** can be electrically connected by conductive epoxy **1340**, **1342**, respectively, on the same surface or side of the PZT. This reduces the number of epoxy bonding steps to two steps and eliminates the difficult to control epoxy height tolerance on the top PZT surface. Also, only a single curing step is needed.

[0065] FIG. **14** is a flow diagram of a manufacturing process using prior epoxy dispensing and bonding steps for attaching the PZT to its suspension as shown in FIG. **3**. Conductive epoxy is first dispensed for grounding (**1410**). Non-conductive epoxy is then dispensed for insulation (**1412**). The PZT is then attached to the suspension, at step **1414**. The epoxies are then cured (**1416**). A second epoxy dispensing step is then performed (**1418**). Finally, that last-dispensed epoxy is cured (**1420**). The process required two separate curing steps.

[0066] FIG. 15 is a flow diagram of a manufacturing process according to the present invention for bonding the PZT to its suspension as shown in FIG. 13. Conductive epoxy is dispensed for grounding (1510). Conductive epoxy is then dispensed for the signal or driving voltage for the PZT (1512). The PZT is then attached to the suspension (1514). Finally, the assembly is cured (1516). The figures illustrate the simplification that is obtained by using a PZT with a wrap-around electrode according to the invention.

[0067] FIGS. 19-21 illustrate the preparation and placement of a PZT having B-staged adhesive according to the invention. FIG. 19 is a perspective view of a PZT 1930 with adhesive 1920, 1924 being applied to what will be the bottom surface of PZT 1930. In general, the adhesive can be applied by any one of a number of known techniques including screen printing, stencil printing, ink jet printing, spraying, application as a film, and others. In general, any patterns desired of a combination of conductive adhesive and/or non-conductive adhesive may be applied to PZT 1930. In the illustrative embodiment shown, the adhesive is sprayed by ink jet heads 1910, 1911, to produce one strip of conductive epoxy 1924, and one strip of non-conductive epoxy 1920. The two strips can be of different thicknesses, with one strip being thicker than the other.

[0068] FIG. 20 is a top perspective view of PZT 1930 with the adhesives being B-staged. In the figure, the epoxy strips 1920, 1924 are being UV B-staged. In general, adhesives 1920, 1924 can be B-staged using any known technique including thermal B-staging and UV B-staging. Furthermore, the two strips of adhesive 1920, 1924 can be B-staged to different extents, leaving one strip more cured or hardened than the other. For UV curing, masks or screens can be used in order to irradiate one strip more than the other.

[0069] FIG. 21 is a side cutaway view of PZT 1930 after it has been applied to a suspension. FIG. 21 is analogous to FIG. 3 and illustrates how the present invention can simplify the prior process. After PZT 1930 has been positioned, pressure is applied to at least the left side of the PZT as seen in the figure in order to squeeze non-conductive epoxy 1920 so that it flows into the gap between the PZT and polyimide layer 314, and covers the previously exposed stainless steel 312 adjacent the PZT. The pressure could be applied by mechanically pressing on the PZT. Alternatively, depending on how much hardening occurred during the B-staging process, the weight of the PZT itself may provide sufficient force and pressure to squeeze non-conductive epoxy 1920 into that gap. After the PZT is positioned, and pressed down if necessary, conductive epoxy 322 is applied so as to bridge the gap between the metallized top surface of the PZT, which defines the drive electrode, and copper contact pad 316 which supplies the drive voltage to the PZT. All of the epoxy in the assembly is then cured at the same time in a single curing step. This process eliminates the need for a second curing step as was required for the assembly and process of FIGS. 3-6.

[0070] FIG. 21 represents only one of a number of possible different types of electrical connections and electrical connection methods for providing the PZT driving voltage and ground to the PZT. Many other connection types and methods are possible, such as disclosed in U.S. Pat. No. 8,189,301 to Schreiber, and U.S. Pat. No. 7,751,153 to Kulangara. The present invention is applicable in suspensions that employ various types of electrical connections to the PZTs.

[0071] As an alternative to the bonding structure shown in FIG. 21, PZT 1930 could be extended farther to the left in the figure, such that the PZT is bonded directly to copper contact pad 316.

[0072] FIGS. 16A-16G illustrate the manufacturing steps for making the PZT with wrap-around electrode shown in FIG. 13. First, a PZT block or wafer 1630 is placed onto a transfer tape 1602, with PZT bottom surface 1631 facing downward (FIG. 16A). Next, an appropriate mask is placed over the top surface of the PZT wafer, and a metallization layer 1604 such as aluminum metallization is sputtered onto the top surface (FIG. 16B). The mask leaves strips 1606 of PZT surface not metallized. Kerf 1608 is then cut into the PZT wafer to separate a first portion, which will be referred to as a PZT precursor 1632, from the rest of the wafer (FIG. 16C). A second mask 1612 is placed over PZT precursor 1632, and additional metallization 1605 is sputtered into the kerfs onto the sides of PZT precursors 1632 within kerfs 1608 in order to make those sides

electrically conductive and electrically continuous with the metallizations on the top surface of the PZT adjacent the kerfs (FIG. 16D).

[0073] Next, the PZT precursor **1632** is flipped over and preferably placed onto a second transfer tape in order to expose what had been the bottom surface **1631** of the PZT precursor (FIG. 16E). That bottom surface **1631** will continue to be referred to as the bottom surface even though it is now facing upward. A metallization layer **1614** is then sputtered onto the entire bottom surface **1631** of the PZT precursor (FIG. 16F). Finally, a new cut **1616** is made into the PZT, separating the first PZT generally in half and defining what will be referred to as first PZT **1634** and second PZT **1636** (FIG. 16G).

[0074] The result of this process is two PZTs **1634**, **1636** each of which has the same structure. A narrow stripe of metallization **1650** on the first PZT's top surface **1633** and near its end, defines a first electrode. The first electrode **1650** electrically wraps around via the metallized side surface **1605** of the PZT to the bottom surface **1631** of the PZT and to the metallization **1604** that generally covers bottom surface **1631**. A second electrode **1652** on the top surface **1633** of the first PZT covers most, but not all, of the PZT top surface **1633**. In this way, a first PZT has been constructed whose first electrode **1650** is located on the same surface of the first PZT as the second electrode **1652**. Generally speaking, the first electrode can be the electrode at which the PZT drive voltage is applied with the second electrode being the electrode at which the PZT is grounded, or vice versa. The second PZT is substantially identical to the first PZT.

[0075] FIGS. 17A and 17B are isometric views of a PZT wafer at selected steps during the manufacturing process according to the present invention. FIG. 17A is an isometric view of the PZT strips **1634**, **1636** at the end of the process in FIG. 16. Now, the PZT is ready for poling; the final dicing operation can be performed afterward to create the individual PZTs as shown in FIG. 17B. The result is a first row of PZTs **1638** and a second row of PZTs **1640**, both having wrap-around electrodes.

[0076] FIGS. 18A-18G illustrate alternative manufacturing steps for making the PZT having a wrap-around electrode. The PZT wafer **1830** is placed onto a transfer tape **1802** with its bottom surface **1831** facing downward (FIG. 18A). Kerfs **1808** are then cut into PZT **1830** to separate a first portion, which will be referred to as a PZT precursor **1832**, from the rest of the wafer (FIG. 18B). The kerfs **1808** on either side of PZT precursor **1832** are then filled with a conductive and hardenable material **1820**, such as conductive epoxy paste containing copper, silver, or other conductive material or particles (FIG. 18C). Silver epoxy is one such commonly used material, and will be used as an example. Silver epoxy **1820** is then allowed to harden. Next, an appropriate mask is placed over the top surface of PZT precursor **1832**, and a metallization layer **1804** such as aluminum metallization is sputtered onto the top surface (FIG. 18D). The mask prevents metallization along two narrow strips **1806**, **1807** on the top surface of the PZT precursor near the ends thereof. The result is that the PZT precursor **1832** has two relatively narrow stripes of metallization **1842**, **1844** on the top surface of the PZT precursor near either end thereof, and a large metallization area **1846** generally centered on the PZT precursor. At this point in the process, each stripe of metallization **1842**, **1844** near an end of the PZT precursor is electrically connected to the silver epoxy **1820** on the end of the PZT precursor to which that stripe is adjacent, and each of the two stripes of metallization **1842**, **1844** and the large metallized region **1846** in the center are all electrically isolated or electrically discontinuous from each other.

[0077] Next, the PZT precursor is flipped over and preferably placed onto a second transfer tape in order to expose what had been the bottom surface **1831** of the PZT precursor (FIG. 18E). That bottom surface **1831** will continue to be referred to as the bottom surface even though it is now facing upward. A metallization layer **1814** is then sputtered onto the entire bottom surface of the PZT precursor (FIG. 18F). Finally, a new cut **1818** is made into the PZT precursor, cutting the PZT precursor generally in half, and another cut **1816** is made through the silver epoxy, thereby separating the PZT precursor in half and defining what will be referred to as first and second PZTs

(FIG. 18G). The cut **1816** is within kerf **1808** and is narrower than kerf **1808** so as to leave the side of the PZT precursor covered with conductive silver epoxy **1820**, and thus serves as an electrical bridge or wrap-around from the PZT's top surface to its bottom surface.

[0078] The result of this process is two PZTs each of which has the same structure. A narrow stripe of metallization **1844** on the first PZT's top surface **1833** and near its end, defines a first electrode. The first electrode **1844** electrically wraps around via the silver epoxy **1820** to bottom surface **1831** of the PZT and to the metallization that generally covers bottom surface **1831**. A second electrode **1852** on the top surface **1833** of the first PZT covers most, but not all, of the PZT top surface **1833**. In this way, a first PZT has been constructed whose first electrode **1844** is located on the same surface of the first PZT as the opposite electrode **1852**. The second PZT is substantially identical to the first PZT.

[0079] FIG. 22 is a side cutaway view of the PZT having a wrap-around electrode such as the PZTs of either FIG. 16G or FIG. 18G, after it has been applied to a suspension. PZT **1830** has a first and wrap-around electrode **1844** which serves as the drive or positive electrode, and a second electrode **1852** which serves as the ground electrode. Drive electrode **1844** is bonded to copper contact pad **316** on polyimide layer **314** over stainless steel substrate **312**, via conductive epoxy **1924**. Ground electrode **1852** is grounded to stainless steel substrate **328** via gold bond pad **326** and conductive epoxy **1928**. Strips of conductive epoxy **1924** and **1928** can be B-staged epoxies as discussed above, with the epoxies fully cured after the parts have been assembled as shown in the figure. This embodiment completely eliminates the need to dispense any epoxy within the suspension assembly room, and eliminates any liquid or paste epoxy from that room, and thereby helps to keep that environment free from contamination.

[0080] Some embodiments described herein are directed to a system for printing conductive and non-conductive adhesives onto a flexible circuit is provided. The system includes a machine base, a flexible circuit positioned on the machine base, and a printing device. The printing device includes an aperture. The aperture is positioned over a predetermined location on the flexible circuit where adhesive material is to be deposited. The system also includes an application device configured to move in a shearing direction about the printing device and deposit the adhesive material into the aperture.

[0081] There are drawbacks to the prior methods of bonding PZTs to suspensions. It can be difficult to control exactly how much epoxy is dispensed, where the adhesive ends up due to flow of the liquid adhesive, and other issues. Various solutions have been proposed that involve, for example, channels underneath the PZTs to control the flow of adhesive and to channel any excess liquid epoxy away from sensitive areas.

[0082] For example, an adhesive attachment can be implemented that has one or more reliefs under or partially under or adjacent to a PZT transducer to control the flow of adhesive by limiting or influencing adhesive travel or flow and simultaneously preventing excessive adhesive fillet height adjacent the piezoelectric motor. Additionally, if the PZT is located at or near the gimbal which carries the magnetoresistive read/write head, it becomes critical to be able to predict and control the flow of adhesive because differences in adhesive flow and distribution from one part to another can adversely affect the geometries, mechanical properties, and resulting performance of the suspension. These issues are particularly pronounced when the PZT is located at a particularly sensitive part of the suspension such as near or at the gimbaled head slider. Repeatability and predictability are especially critical in that area. Still further, the presence of liquid epoxy and its dispensing equipment within the final assembly room represents both a potential source of contamination, as well as an additional and expensive manufacturing step.

[0083] FIG. 23 is a perspective view showing the suspension **10**, in accordance with an embodiment of the disclosure. The suspension **10** can be supported by a load beam **23**. The suspension is generally comprised of various suspension components, and for example can include a flexure **22**. The flexure **22** supports a head gimbal assembly **21**. A slider, which constitutes a

read/write head, is provided on the head gimbal assembly **21**. In some embodiments, the slider includes magnetoresistive (MR) elements, which are capable of conversion between magnetic and electrical signals. The MR elements serve to access data, that is, write or read data to or from the disk.

[0084] The head gimbal assembly **21** includes microactuator elements **31** and **32**. Microactuator elements **31** and **32**, according to some embodiments, are formed of piezoelectric plates of lead zirconate titanate (“PZT”) or the like. The microactuator elements **31** and **32** have the function of pivoting the slider in the sway direction by means of a structure. The suspension **10** can be configured as a multi-stage actuator, such as for example a dual-stage-actuator (DSA) type, meaning two microactuator elements **31** and **32** mounted in the head gimbal assembly **21** or a tri-stage (as shown) including one or more microactuators **25** mounted in a region of the metal base **40**. It should be understood, any configurations of microactuator elements may be implemented herein.

[0085] The head gimbal assembly **21** includes a metal base **40**. In some embodiments, the metal base **40** is formed of a stainless-steel plate. The flexure **22** also includes a conductive circuit portion including one or more conductors, such as traces. The conductive circuit portion includes a conductor that connects to the slider. The conductor can also connect to electrodes of the microactuator elements **31** and **32**. Conductive and non-conductive adhesives (not shown) are used to connect the microactuator elements **31** and **32** to the flexure **22**.

[0086] The conductive and non-conductive adhesives are typically applied to the microactuator elements **31** and **32** or flexure **22** using syringe dispensing methods. For example, a syringe can be positioned over desired locations and air pressure or pneumatic mechanisms extrude adhesive from the syringe onto the microactuator elements **31** and **32** or flexure **22**. Dots of adhesive are deposited one after another onto the microactuator elements **31** and **32** or flexure **22**.

[0087] It is often difficult to control the volume of adhesive applied using the syringe dispensing methods due to the size deposits. Making the deposits using syringe dispensing methods is inefficient, and often times inaccurate. Accordingly, new improvements are needed.

[0088] FIG. **24** illustrates a system **2400** for screen printing conductive and non-conductive adhesives onto various components and/or flexible circuits of the suspension **10** according to embodiments of the present disclosure. The system **2400** can include a machine base **2420**, a panel suspension components or flexible circuits **2440**, a screen or stencil **2450**, and a squeegee **2460**. The screen or stencil **2450** can include apertures **2451**. The apertures **2451** can be positioned over the suspension components or flexible circuits **2440** at a desired position for printing or depositing the conductive and non-conductive adhesive onto the suspension component or flexible circuit. The squeegee **2460** is configured to move in a shearing direction **2490** with respect to the screen or stencil **2450**. In this example, the conductive and non-conductive adhesives are used to connect the microactuator elements of FIG. **23** to the circuit traces on the flexure of FIG. **23**.

[0089] In another example, the adhesives produced by the system and methods described herein are used to provide various features (for example as shown in FIGS. **26-28**, described below) on suspension components, such as for example on a baseplate **40** or integrated baseplate **40** and loadbeam **23**. These adhesives may be useful for various purposes such as providing structural benefits such as increasing the torsional stiffness, among other purposes and advantages.

[0090] Referring again to FIG. **24**, as the squeegee **2460** moves in the shearing direction **2490**, the squeegee **2460** can deposit adhesive material **2470** within apertures **2451** of the screen or stencil **2450**. The adhesive material **2470** can be screen printed in the form of a viscous fluid. Once the adhesive material **2470** is deposited within the apertures **2451** of the screen or stencil **2450**, the screen or stencil **2450** will be removed, and the microactuator elements **31** and **32** will be positioned over the screened adhesive material **2470** at a desired position. Then, the panel of flexible circuits or suspension components **2440** will undergo heat and/or UV exposure. In one example the heat and/or UV exposure will cure the adhesive material **2470** to a solid state and

permanently connect the microactuator elements of FIG. 23 to the circuit traces on the flexure of FIG. 23. In another example, the heat and/or UV exposure will partially cure or B-stage the adhesive material **2470** to a semi-solid state to form features on the baseplate **40** or an integrated baseplate **40** and load beam **23**.

[0091] FIGS. 25A and 25B are a process flow diagrams illustrative methods for screen printing conductive and non-conductive adhesives onto flexible circuits and suspension components, respectively, according to two examples. Referring to FIG. 25A, the method **400** is described with reference to the elements described in FIG. 24. At step **401**, the panel of flexible circuits **2440** is loaded onto the machine base **2420**. At step **402**, the screen or stencil **2450** can be placed over the panel of flexible circuits **2440**. At step **403**, adhesive material **2470** can be applied to the screen or stencil **2450** at a first location. The first location can be a distal location. At step **404**, the squeegee **2460** can be positioned at the distal location with the adhesive material **2470**. At step **405**, the squeegee **2460** can move across the screen or stencil **2450** in a proximal direction, away from the distal location. In the process of moving the squeegee across the screen or stencil **2450**, the adhesive material **2470** is deposited within apertures **2451** of the screen or stencil **2450**.

[0092] FIG. 25B is a process flow diagram showing methods for screen printing conductive and non-conductive adhesives onto suspension components according to another example. The system shown in FIG. 24 may also be used to carry out this process. The method **500** will be described with reference to the elements described in FIG. 24. At step **501**, one or more suspension components, or a panel of suspension components, **2440** is loaded onto the machine base **2420**. At step **502**, the screen or stencil **2450** can be placed over the suspension component(s) or panel **2440**. At step **503**, adhesive material **2470** can be applied to the screen or stencil **2450** at a first location. The first location can be a distal location. At step **504**, the squeegee **2460** can be positioned at the distal location with the adhesive material **2470**. At step **505**, the squeegee **2460** can move across the screen or stencil **2450** in a proximal direction, away from the distal location. In the process of moving the squeegee across the screen or stencil **2450**, the adhesive material **2470** is deposited within one or more apertures **2451** of the screen or stencil **2450**. The one or more apertures is sized to receive a desired volume of adhesive material. In some embodiments, the one or more apertures has a diameter in the range of 0.5 mm to 5 mm, and 2.5 mm to 3.0 mm.

[0093] The configurations shown and described above allow for rapid deposition of adhesive material, as the system **2400** is able to perform the deposits simultaneously. Furthermore, the system **2400** allows for better control of the deposit volume that is much tighter than syringe deposition. More precise volume control is achieved, because the deposit volume is precisely controlled by the apertures **2451** in the printed location and the squeegee **2460**, which sweeps the fluid across the apertures **2451**. It should be understood that while the present disclosure details screen or stencil printing viscous liquid adhesives, other printing methods such as flexographic printing or gravure printing can be implemented herein.

[0094] FIGS. 26A and 26B are top and perspective views, respectively, showing a printed adhesive formed on suspension components such as a stiffener or baseplate according to an example of the method shown in FIG. 25B. In this example, the suspension component is a stiffener or baseplate **602**. A printed adhesive **604** is printed onto the baseplate **602** at a desired location according to the method steps described above and shown in FIG. 25B. The printed adhesive **604** can be of any shape, size and location on the suspension component.

[0095] FIGS. 27A and 27B illustrate another example of a suspension component with a printed adhesive formed according to methods described in the present disclosure. In this example, the suspension component is an integrated baseplate and load beam assembly **700** which is comprised of a baseplate **702a** and load beam **702b**. A printed adhesive **704** is printed onto the baseplate **702a** at a desired location according to the method steps described above and shown in FIG. 25B. The printed adhesive **704** can be of any shape, size and location on the suspension component. In this example, the printed adhesive **704** is printed around the circumference of an opening with hub

geometry **706**. The opening and hub **706** can be formed using known manufacturing methods in the field.

[0096] FIG. **28** shows another example of a suspension component with a printed adhesive formed according to methods described in the present disclosure. In this example, as suspension assembly **800** is shown having suspension components comprised of an integrated baseplate and load beam assembly **802** and a suspension arm **804**. The integrated baseplate and load beam assembly **802** includes a baseplate **806a** and load beam **806b**. A first printed adhesive **808** is printed onto the baseplate **802a** at a desired location according to the method steps described above and shown in FIG. **25B**. The first printed adhesive **808** can be of any shape, size and location on the suspension component. In this example, the printed adhesive **808** is printed around the circumference of an opening with hub geometry, at a proximal end of the baseplate **806a**. The opening and hub **810** can be formed using known manufacturing methods in the field.

[0097] In this example, a second printed adhesive **812** is formed on the suspension assembly using the method steps described above and shown in FIG. **25B**. Of particular advantage, the second adhesive **812** is formed on the baseplate **806b** at a location on the baseplate that imparts a structural effect on the suspension component. In this example the second adhesive increases the torsional stiffness of the suspension **800** when the arm **804** is attached. In this example, the second adhesive **812** is located at a distal end of the baseplate **806a** and has an area that is greater than the area of the first adhesive **808**. The second printed adhesive **812** can be of any shape, size and location on the suspension component, and is preferably configured to provide a structural benefit to the suspension assembly, such as to increase torsional stiffness, among other structural benefits.

[0098] It will be understood that the terms “generally,” “approximately,” “about,” and “substantially,” as used within the specification and the claims herein allow for a certain amount of variation from any exact dimensions, measurements, and arrangements, and that those terms should be understood within the context of the description and operation of the invention as disclosed herein.

[0099] It will further be understood that terms such as “top,” “bottom,” “above,” and “below” as used within the specification and the claims herein are terms of convenience that denote the spatial relationships of parts relative to each other rather than to any specific spatial or gravitational orientation. Thus, the terms are intended to encompass an assembly of component parts regardless of whether the assembly is oriented in the particular orientation shown in the drawings and described in the specification, upside down from that orientation, or any other rotational variation.

[0100] All features disclosed in the specification, including the claims, abstract, and drawings, and all the steps in any method or process disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. Each feature disclosed in the specification, including the claims, abstract, and drawings, can be replaced by alternative features serving the same, equivalent, or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

[0101] It will be appreciated that the term “present invention” as used herein should not be construed to mean that only a single invention having a single essential element or group of elements is presented. Similarly, it will also be appreciated that the term “present invention” encompasses a number of separate innovations which can each be considered separate inventions. Although the present invention has thus been described in detail with regard to the preferred embodiments and drawings thereof, it should be apparent to those skilled in the art that various adaptations and modifications of the present invention may be accomplished without departing from the spirit and the scope of the invention. For example, instead of selectively applying and partially curing adhesive on the PZT, adhesively could be selectively applied and partially cured on other suspension components such as the flexure. Accordingly, it is to be understood that the

detailed description and the accompanying drawings as set forth hereinabove are not intended to limit the breadth of the present invention.

Claims

1. A system for printing conductive and non-conductive adhesives onto one or more suspension components, the system comprising: a machine base configured to receive at least one suspension component; a printing device including at least one aperture, the at least one aperture is configured to be positioned over a location on the at least one suspension component where adhesive material is to be deposited; an application device configured to move in a shearing direction about the printing device and deposit the adhesive material into the at least one aperture.
2. The system of claim 1, wherein the machine base is configured to receive a panel of more than one suspension component, wherein each of the suspension components includes at least one location where the adhesive material is to be deposited.
3. The system of claim 1, wherein the printing device includes either a screen or a stencil.
4. The system of claim 1, wherein the adhesive material includes conductive adhesive material.
5. The system of claim 1, wherein the adhesive material includes non-conductive adhesive material.
6. The system of claim 1, wherein the adhesive material is a viscous fluid.
7. The system of claim 1, further comprising a heat source configured to cure the adhesive material deposited into the at least one aperture.
8. The system of claim 1, further comprising a UV light source configured to cure the adhesive material deposited into the at least one aperture.
9. The system of claim 1, wherein the at least one aperture is sized to receive a desired volume of adhesive material.
10. The system of claim 9, wherein the at least one aperture has a diameter in the range of about 0.5 mm to 5.0 mm.
11. The system of claim 9, wherein the at least one aperture has a diameter in the range of about 2.5 mm to 3.0 mm.
12. The system of claim 1, wherein the adhesive is deposited on the suspension component at a location that imparts a structural effect on the suspension component.
13. The system of claim 12, wherein the adhesive deposited on the suspension component increases the torsional stiffness of the suspension component.
14. A method of printing adhesive on a suspension component, comprising the steps of: loading one or more suspension components onto a machine base; placing a screen or stencil over the one or more suspension components, the screen or stencil including one or more apertures formed therein; applying adhesive material to the screen or stencil at a first location; positioning a squeegee at the first location with the adhesive material; and moving the squeegee across the screen or stencil to deposit the adhesive material into the one or more apertures.
15. The method of claim 14, further comprising: partially curing the adhesive material deposited into the at least one aperture.
16. The method of claim 14, wherein the loading step further comprises loading a panel comprising a plurality of suspension components, and each of the suspension components in the panel includes at least one location where the adhesive material is to be deposited.
17. The method of claim 14, wherein the first location is a distal location, and the squeegee moves across the screen or stencil in a proximal direction.
18. The method of claim 14, wherein the adhesive material includes conductive adhesive material.
19. The method of claim 14, wherein the adhesive material includes non-conductive adhesive material.
20. The method of claim 14, wherein the adhesive material is a viscous fluid.
21. The method of claim 14, wherein the one or more apertures is sized to receive a desired volume

of adhesive material.

22. The method of claim 21, wherein the one or more apertures has a diameter in the range of about 0.5 mm to 5.0 mm.

23. The method of claim 21, wherein the one or more apertures has a diameter in the range of about 2.5 mm to 3.0 mm.

24. The method of claim 14, wherein the adhesive is deposited on the suspension component at a location that imparts a structural effect on the suspension component.

25. The method of claim 24, wherein the adhesive deposited on the suspension component increases the torsional stiffness of the suspension component.
